ELASTICITY — ASSIGNMENT 01

Single Correct · Multiple Correct · Subjective

SECTION A — Single Correct Type Q1–9

Only one option is correct.

1. The maximum load a wire can withstand without breaking, when its length is reduced to half of its original length, will

(A) be double. (B) be half.
(C) be four times. (D) remain same.

2. The temperature of a wire is doubled. The Young's modulus of elasticity

(A) will also double.
(B) will become four times.
(C) will remain same.
(D) will decrease.

3. A spring is stretched by applying a load to its free end. The strain produced in the spring is

(A) volumetric.
(B) shear.
(C) longitudinal and shear.
(D) longitudinal.

4. Overall changes in volume and radii of a uniform cylindrical steel wire are 0.2% and 0.002% respectively when subjected to some suitable force. Longitudinal tensile stress acting on the wire is ($Y = 2.0 \times 10^{11}\ \mathrm{N\,m^{-2}}$)

(A) $3.2 \times 10^{9}\ \mathrm{N\,m^{-2}}$ (B) $3.2 \times 10^{7}\ \mathrm{N\,m^{-2}}$
(C) $3.6 \times 10^{9}\ \mathrm{N\,m^{-2}}$ (D) $3.9 \times 10^{8}\ \mathrm{N\,m^{-2}}$

5. A solid sphere of radius R made of material of bulk modulus K is surrounded by a liquid in a cylindrical container. A massless piston of area A floats on the surface of the liquid. When a mass m is placed on the piston to compress the liquid, the fractional change in the radius of the sphere $\delta R/R$ is

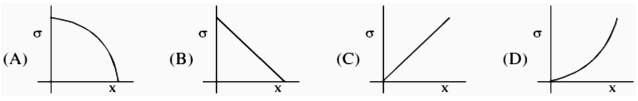
(A) mg/AK (B) $mg/3AK$ (C) mg/A (D) $mg/3AR$

6. A wire suspended vertically from one of its ends is stretched by attaching a weight of 200 N to the lower end. The weight stretches the wire by 1 mm. Then the elastic energy stored in the wire is

[AIEEE 2003]

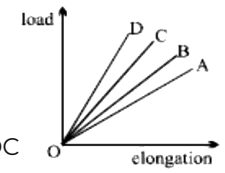
(A) 0.2 J (B) 10 J (C) 20 J (D) 0.1 J

7. A uniform rod is rotating in a gravity-free region with a certain constant angular velocity. The variation of tensile stress with distance x from the axis of rotation is best represented by which of the following graphs?

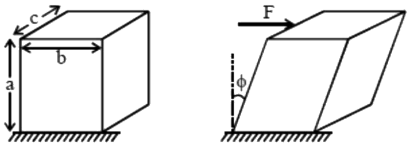


8. The load versus strain graph for four wires of the same material is shown in the figure. The thickest wire is represented by the line

(A) OB (B) OA (C) OD (D) OC



9. A cuboidal block of sides a, b and c is fixed on the ground. The top is pushed by a horizontal force F as shown. The angle ϕ by which the block deforms is (η is modulus of rigidity)



(A) $\dfrac{F}{ab\eta}$ (B) $\dfrac{F}{ac\eta}$ (C) $\dfrac{F}{bc\eta}$ (D) $\dfrac{F}{\sqrt{b^2+c^2}\,\eta}$

SECTION B — Multiple Correct Type Q10–17

One or more options may be correct.

10. A wire is suspended from the ceiling and stretched under the action of a weight F suspended from its other end. The force exerted by the ceiling on it is equal and opposite to the weight.

(A) Tensile stress at any cross section A of the wire is F/A.
(B) Tensile stress at any cross section is zero.
(C) Tensile stress at any cross section A of the wire is $2F/A$.
(D) Tension at any cross section A of the wire is F.

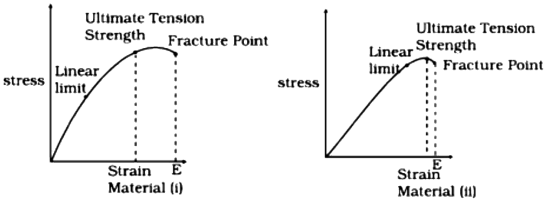
11. A copper and a steel wire of the same diameter are connected end to end. A deforming force F is applied to this composite wire which causes a total elongation of 1 cm. The two wires will have

(A) the same stress. (B) different stress.
(C) the same strain. (D) different strain.

12. A body of mass M is attached to the lower end of a metal wire, whose upper end is fixed. The elongation of the wire is l.

(A) Loss in gravitational potential energy of M is Mgl
(B) The elastic potential energy stored in the wire is Mgl
(C) The elastic potential energy stored in the wire is $\frac{1}{2}Mgl$
(D) Heat produced is $\frac{1}{2}Mgl$.

13. The stress–strain graphs for two materials are shown in the figure (assume same scale).

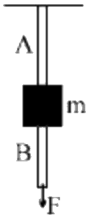


(A) Material (ii) is more elastic than material (i) and hence material (ii) is more brittle.
(B) Material (i) and (ii) have the same elasticity and the same brittleness.
(C) Material (ii) is elastic over a larger region of strain as compared to (i).
(D) Material (ii) is more brittle than material (i).

14. A composite rod consists of a steel rod of length 25 cm and area $2A$ and a copper rod of length 50 cm and area A. The composite rod is subjected to an axial load F. The Young's moduli of steel and copper are in the ratio 2 : 1.

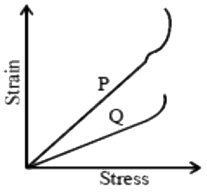
(A) the extension produced in copper rod will be more.
(B) the extension in copper and steel parts will be in the ratio 2 : 1.
(C) the stress applied to the copper rod will be more.
(D) no extension will be produced in the steel rod.

15. The wires A and B shown in the figure are made of the same material and have radii r_A and r_B respectively. The block between them has a mass m. When the force F is $mg/3$, one of the wires breaks.



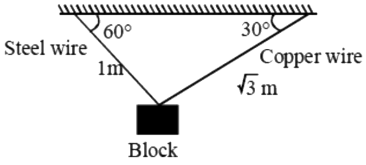
(A) A breaks if $r_A = r_B$
(B) A breaks if $r_A < 2r_B$
(C) Either A or B may break if $r_A = 2r_B$
(D) The lengths of A and B must be known to predict which wire will break

16. In plotting stress versus strain curves for two materials P and Q, a student by mistake puts strain on the y-axis and stress on the x-axis as shown in the figure. Then the correct statement(s) is (are) **[JEE Advanced 2015]**



(A) P has more tensile strength than Q
(B) P is more ductile than Q
(C) P is more brittle than Q
(D) The Young's modulus of P is more than that of Q

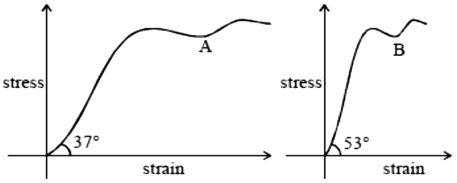
17. A block of weight 100 N is suspended by copper and steel wires of the same cross-sectional area 0.5 cm^2 and lengths $\sqrt{3}$ m and 1 m respectively. Their other ends are fixed on a ceiling as shown in the figure. The angles subtended by the copper and steel wires with the ceiling are 30° and 60° respectively. If the elongation in the copper wire is Δl_C and in the steel wire is Δl_S, then the ratio $\Delta l_C/\Delta l_S$ is ____. (Young's modulus for copper and steel: 1×10^{11} N/m^2 and 2×10^{11} N/m^2 respectively.) **[JEE Advanced 2019]**



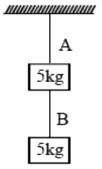
SECTION C — Subjective Questions Q18–24

18. A steel wire of length 4.5 m and a copper wire of length 3.5 m are stretched by the same amount under a given load. If the ratio of the Young's moduli of steel to that of copper is $\frac{12}{7}$, what is the ratio of the cross-sectional area of the steel wire to the copper wire?

19. The diagram shows stress–strain graphs for two materials A and B. The graphs are drawn to scale. Find the ratio of the Young's moduli of material A to material B.

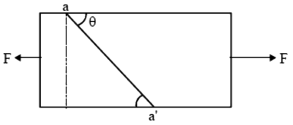


20. Two identical wires A and B of the same material are loaded as shown in the figure. If the elongation in wire B is 1.5 mm, what is the elongation in A? (Masses of A and B can be neglected.)



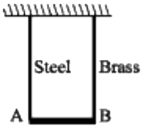
21. A wire of length L and radius r is clamped rigidly at one end. When the other end of the wire is pulled by a force f, its length increases by l. Another wire of the same material, of length $2L$ and radius $2r$, is pulled by a force $2f$. Find the increase in length of this wire.

22. Consider a long steel bar under a tensile stress due to forces F acting at the edges along the length of the bar (see figure).



Consider a plane making an angle θ with the length. What are the tensile and shearing stresses on this plane? (a) For what angle is the tensile stress a maximum? (b) For what angle is the shearing stress a maximum?

23. A light rigid bar AB is suspended horizontally from two vertical wires, one of steel and one of brass, as shown in the figure. Each wire is 2.00 m long. The diameter of the steel wire is 0.60 mm and the length of the bar AB is 0.20 m. When a mass of 10 kg is suspended from the centre of AB, the bar remains horizontal. (i) What is the tension in each wire? (ii) Calculate the extension of the steel wire and the energy stored in it. (iii) Calculate the diameter of the brass wire. (iv) If the brass wire were replaced by another brass wire of diameter 1 mm, where should the mass be suspended so that AB would remain horizontal? Young's modulus for steel $= 2.0 \times 10^{11}$ Pa; for brass $= 1.0 \times 10^{11}$ Pa.



24. A wire of cross-sectional area 4×10^{-4} m^2, modulus of elasticity 2×10^{11} N/m^2 and length 1 m is stretched between two vertical rigid poles. A mass of 1 kg is suspended at its middle. Calculate the angle it makes with the horizontal.